

AI2: Intro to Information Theory

Information Theoretic Quantities

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Outline

By the end of this lecture you will learn:

- What is information theory, and why we study it in this module
- Shannon's Self-information and Entropy
- Joint Entropy
- Conditional Entropy
- Relative Entropy (also called Kullback-Leibler divergence)
- Mutual Information

What is Information Theory?

Information theory answers:

How to quantify information?

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How to quantify information?

Example: Consider the following 2 sentences:

- 1 "Alice is a person."
- 2 "Alice is a student."

Which sentence contains more information? Why?

How can we quantify this?

Information Theory = Quantifying information

- The field of Information Theory was established by Claude Shannon in the 1940's
- Original motivation: Practical solutions to communication problems, e.g. error correcting codes, Audio/Image/Video Compression.

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Why we are interested?

Close connections & applications in Machine Learning^a, Statistics, Engineering.

^aSee David MacKay's book "Information Theory, Inference, and Learning Algorithms"

- We will see shortly that Logistic Regression can be interpreted in terms of information measures.
- In the second lecture we will learn about Decision Trees, and Feature Selection, both of which use information measures.

Example: How to quantify the information content of English?

Consider the following 2 sentences:

- 1 "Alice is a person."
- 2 "Alice is a student."

The 2nd sentence contains more information than the 1st.

Question: Can we have a mathematical measure that tells us how much more information second sentence have as compared to the first?

Answer from Shannon: "the semantic aspects of data are irrelevant", i.e., the nature and meaning of data are not the information *per se*. Instead, he quantified information in a probability distribution as a degree of "uncertainty" or "surprise".

Self Information

To measure the *information content of an event*, Shannon proposed **self-information** based on 3 axioms:

- An event with 100% certainty is perfectly unsurprising and yields no information.
- The less probable an event is, the more surprising it is, so the more information it yields.
- If two independent events are measured separately, the total amount of information is the sum of the self-informations of the individual events.

Self Information

Definition (Self-information of an outcome of a random variable)

Given a random variable X with probability mass function $P_X(x)$, the self-information of measuring X as outcome x is defined as:

$$I_X(x) = -\log_b[P_X(x)] = \log_b \frac{1}{P_X(x)}$$

Self-information quantifies the level of surprise in observing a particular outcome x for a random variable X that has PMF $P_X(x)$.

Units: The base b of the logarithm determines the units of self-information:

- $b = 2$: Measured in *bits*.
- $b = e$: Measured in *natural units* or *nats*.
- $b = 10$: Measured in *dits*, *bans*, or *hartleys*.

Self-Information and Log-Odds

- Self-information and logit (log-odds) are related concepts.
- For an event A , occurring with probability p (hence the probability of the event not occurring $P(\neg A) = 1 - p$, the logit function is defined as:

$$\text{logit}(A) = \log\left(\frac{p}{1-p}\right) = \log(p) - \log(1-p)$$

Relationship:

$$\text{logit}(A) = I(\neg A) - I(A)$$

where $I(A)$ and $I(\neg A)$ represent the self-information of events A and $\neg A$ respectively.

We encountered this in Logistic Regression, with A being the value of the target variable given an input.

Entropy

Entropy quantifies the uncertainty in a random variable X .

Given a **discrete**¹ **random variable** X with range $R_X = \{x_1, x_2, \dots, x_m\}$ and probability mass function $P_X(x)$, the entropy of X is defined as:

Definition (Entropy)

$$\begin{aligned} H(X) &\equiv E[I_X(X)] \equiv - \sum_{i=1}^m \mathbb{P}(X = x_i) \log_b \mathbb{P}(X = x_i) \\ &\equiv E \left[\log_b \frac{1}{P_X(X)} \right] \equiv -E[\log_b P_X(X)] \end{aligned}$$

Notation: When it's clear from the context which random variable we are talking about, we may write $P(x)$ for $P_X(x)$, and likewise $I(x)$ for $I_X(x)$.

¹For continuous random variables, **differential entropy** is used instead.

Example 1: Fair and Biased Coins

Question: Denote the outcome of tossing a fair coin as X and a biased coin (with heads probability 0.7) as Y . Calculate their entropies and interpret the results.

Solution:

- **Fair Coin:**

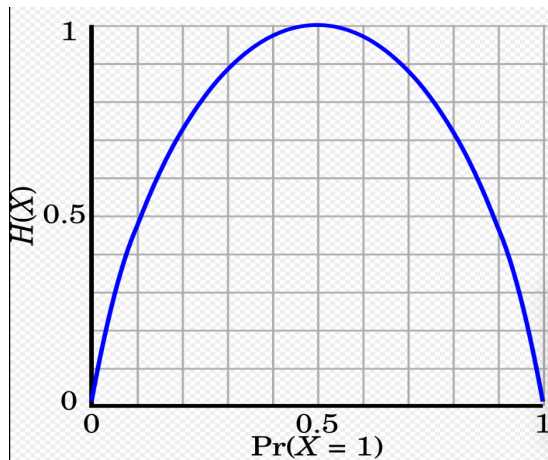
$$H(X) = - \sum_{i=1}^2 P(x_i) \log_2 P(x_i) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) = 1 \text{ bit}$$

- **Biased Coin:**

$$H(Y) = - (0.7 \log_2 0.7 + 0.3 \log_2 0.3) \approx 0.8816 \text{ bits}$$

Interpretation: The biased coin has lower entropy since its outcome is more predictable compared to a fair coin.

Entropy for Bernoulli distribution



Next up: Joint Entropy and Conditional Entropy

Notations refresher

We will restrict ourselves to discrete random variables for simplicity. Let X, Y two discrete random variables.

- The joint entropy and conditional entropy are based on joint PMF and conditional PMF, or joint probability and conditional probability.
 - ▶ If you need a refresher, read the following first: *Introduction to Probability, Statistics, and Random Processes*.
- Simplified notations we will use:
 - ▶ Marginal PMF of X : $P_X(\cdot) = P(\cdot)$
 - ▶ Marginal probability of " $X = x$ ": $\mathbb{P}(X = x) = P(x)$
 - ▶ Joint PMF: $P_{X,Y}(\cdot, \cdot) = P(\cdot, \cdot)$
 - ▶ Joint PMF at (x, y) : $\mathbb{P}(X = x, Y = y) = P(x, y)$
 - ▶ Conditional PMF: $P_{X|Y}(\cdot|\cdot) = P(\cdot|\cdot)$
 - ▶ Conditional PMF at (x, y) : $\mathbb{P}(X = x|Y = y) = P(x|y)$. Note: This is a deterministic value – but if we write $P(x|Y)$, this is a random variable since Y is a random variable.

Joint Entropy

Definition (Joint Entropy)

- "A measure of the uncertainty associated with a set of variables."

For discrete random variables X and Y :

$$H(X, Y) = -E[\log P(X, Y)] = - \sum_{x \in R_X} \sum_{y \in R_Y} P(x, y) \log P(x, y)$$

Conditional Entropy

Definition (Conditional Entropy)

- "Quantifies uncertainty of the outcome of a random variable Y given the outcome of another random variable X ."

$$H(Y|X) = -E[\log P(Y|X)] = - \sum_{x \in R_X} \sum_{y \in R_Y} P(x, y) \log P(y|x)$$

Note: In practice we don't use this definition directly. Instead, we can re-write this expression (next slide) in a way to avoid having to compute $P(y|x)$ at all.

Chain Rule for Conditional Entropy

By definition of Conditional Entropy, and the Bayes rule identity for conditional probability, $P(y|x) = \frac{P(x,y)}{P(x)}$, we have:

$$\begin{aligned} H(Y|X) &= - \sum_{x \in R_X} \sum_{y \in R_Y} P(x,y) \log P(y|x) \\ &= - \sum_{x \in R_X} \sum_{y \in R_Y} P(x,y) \log \left(\frac{P(x,y)}{P(x)} \right) \\ &= - \sum_{x \in R_X} \sum_{y \in R_Y} P(x,y) [\log P(x,y) - \log P(x)] \\ &= H(X, Y) - H(X) \end{aligned}$$

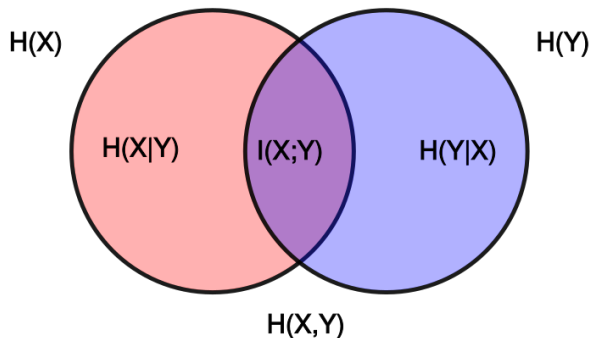
Similarly:

$$H(X|Y) = H(X, Y) - H(Y)$$

Chain rule for conditional entropy – visually

$$H(Y|X) = H(X, Y) - H(X)$$

$$H(X|Y) = H(X, Y) - H(Y)$$



Relative Entropy (Kullback-Leibler Divergence)

KL quantifies the distance between two probability distributions.

Let P and Q be two probability distributions of a discrete random variable X :

$$D_{KL}(P \parallel Q) = \sum_{x \in R_X} P(x) \log \frac{P(x)}{Q(x)} = E \left[\log \frac{P(X)}{Q(X)} \right]$$

By convention:

- $0 \log \frac{0}{Q} = 0$
- $P \log \frac{P}{0} = \infty$

Properties:

- $D_{KL}(P \parallel Q) \geq 0$
- $D_{KL}(P \parallel Q) = 0$ if $P(x) = Q(x)$
- Not symmetric: $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$

Kullback-Leibler Divergence: Examples

For a binary random variable X with range $R_X = \{0, 1\}$, assume two distributions P and Q are defined as:

$$P(0) = 1 - r, \quad P(1) = r$$

$$Q(0) = 1 - s, \quad Q(1) = s$$

KL Divergences:

$$D_{KL}(P \parallel Q) = (1 - r) \log \frac{1 - r}{1 - s} + r \log \frac{r}{s}$$

$$D_{KL}(Q \parallel P) = (1 - s) \log \frac{1 - s}{1 - r} + s \log \frac{s}{r}$$

Examples:

- If $r = s$, then $D_{KL}(P \parallel Q) = D_{KL}(Q \parallel P) = 0$
- If $r = \frac{1}{2}$ and $s = \frac{1}{4}$:

$$D_{KL}(P \parallel Q) = 0.2075$$

$$D_{KL}(Q \parallel P) = 0.1887$$

KL divergence - visualisation

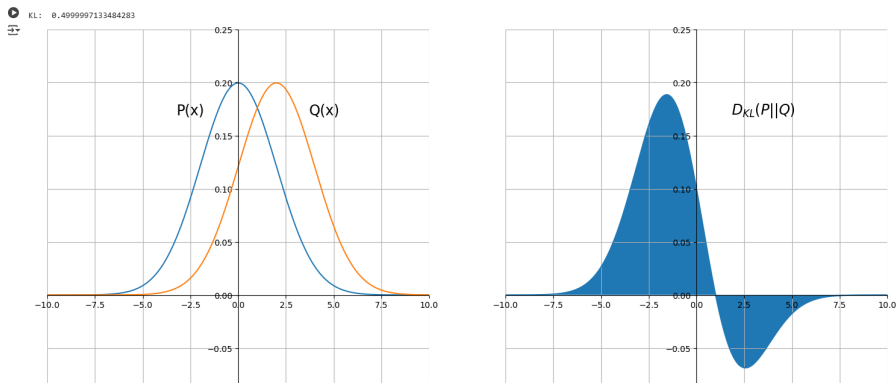


Figure: **Left:** Two Gaussian distributions having means 0 and 2 respectively and identical variance. **Right:** The plotted function is the point-wise contribution to the KL divergence from each possible outcome $x \in [-10, 10]$. The *shaded area* (i.e. a number, in this case 0.499) is the KL divergence between the two Gaussians.

Other divergences and distances between probability distributions

Cross Entropy: For two discrete distributions P and Q , cross entropy is defined as:

$$H(P, Q) = - \sum_{x \in R_X} P(x) \log Q(x) = H(P) + D_{KL}(P \parallel Q)$$

Question: Recall the cost function of logistic regression?

$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \sigma_{\theta}(\mathbf{x}_i) + (1 - y_i) \log(1 - \sigma_{\theta}(\mathbf{x}_i))]$$

What does it correspond to?

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What does it correspond to?

Answer: It's the average *cross-entropy* between Bernoulli distributions! In the i -th term these are $P(1) = y_i$; $P(0) = 1 - y_i$ and $Q(1) = \sigma_{\theta}(\mathbf{x}_i)$; $Q(0) = 1 - \sigma_{\theta}(\mathbf{x}_i)$.

Last week we saw that minimising this amounts to MLE. This example illustrates the tight connection between information theory and statistical estimation.

Mutual Information

Mutual information measures the information that two random variables X and Y share.

Intuition: It quantifies how much knowing one variable reduces uncertainty about the other.

Definition (Mutual Information)

For two discrete random variables X and Y , the mutual information is defined as:

$$I(X; Y) = \sum_{x \in R_X} \sum_{y \in R_Y} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

Mutual Information and KL Divergence

The mutual information $I(X; Y)$ can be expressed as the Kullback-Leibler (KL) divergence between the joint distribution and the product of marginal distributions:

$$I(X; Y) = \sum_{x \in R_X} \sum_{y \in R_Y} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

$$I(X; Y) = D_{KL}(P_{(X,Y)} \parallel P_X P_Y)$$

Interpretation: Mutual information measures the divergence ('error') of modelling the joint probability distribution $P_{(X,Y)}$ as the product of marginals $P_X P_Y$.

Special Case: When X and Y are independent:

$$P(x, y) = P(x)P(y) \implies I(X; Y) = D_{KL}(P_{(X,Y)} \parallel P_X P_Y) = 0$$

Mutual Information, Entropy, and Conditional Entropy

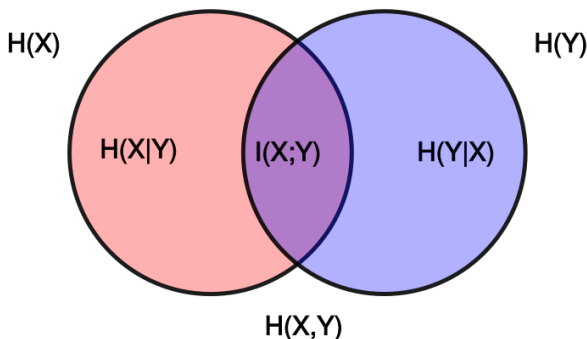
Relationship/ identities (recommended to verify as homework):

$$\begin{aligned} I(X; Y) &\equiv H(X) - H(X|Y) \\ &\equiv H(Y) - H(Y|X) \\ &\equiv H(X) + H(Y) - H(X, Y) \\ &\equiv H(X, Y) - H(Y|X) - H(X|Y) \end{aligned}$$

These equations reveal how mutual information quantifies the reduction in uncertainty about one random variable given knowledge of another.

Visualizing Mutual Information

The identities on the previous slide can be visualised as follows:



(Fig: <https://commons.wikimedia.org/w/index.php?curid=11245361>)

Properties of Mutual Information

Key Properties:

- **Non-negative:** $I(X; Y) \geq 0$
- **Symmetric:** $I(X; Y) = I(Y; X)$
- **Measures statistical dependence:**
 - ▶ $I(X; Y) = 0$ if and only if X and Y are independent.
 - ▶ $I(X; Y)$ increases with the dependence between X and Y and with their individual entropies $H(X)$ and $H(Y)$.
- $I(X; X) = H(X) - H(X|X) = H(X) + 0 = H(X)$

In practice, we use Mutual Information (MI) to decide which features of the data to use for classification. The features that have highest MI with the class label are the informative features.

Question and Further Reading (Optional)

Question: Why should I learn the complex mathematics in this lecture?

Answer: Understanding these concepts will help you comprehend how more advanced Generative AI techniques: such as Stable Diffusion and GPTs (at least GPT 1/2) work (to some extent).

- Stable Diffusion probabilistic models: AI model generating images from text descriptions.
- GPT (Generative Pre-trained Transformer):
 - ▶ GPT1: Improving Language Understanding by Generative Pre-Training
 - ▶ GPT2: Language Models are Unsupervised Multitask Learners
 - ▶ GPT3: Language Models are Few-Shot Learners

Further Reading:

- Diffusion Probabilistic Models: What are Diffusion Models?
- Learn about Diffusion Models
- McKinsey: What is Generative AI?

Recap what we learned

- What is information theory and why we study it in this module
- Shannon's Self-information and Entropy
- Joint Entropy
- Conditional Entropy
- Relative Entropy (also called Kullback-Leibler divergence)
- Mutual Information